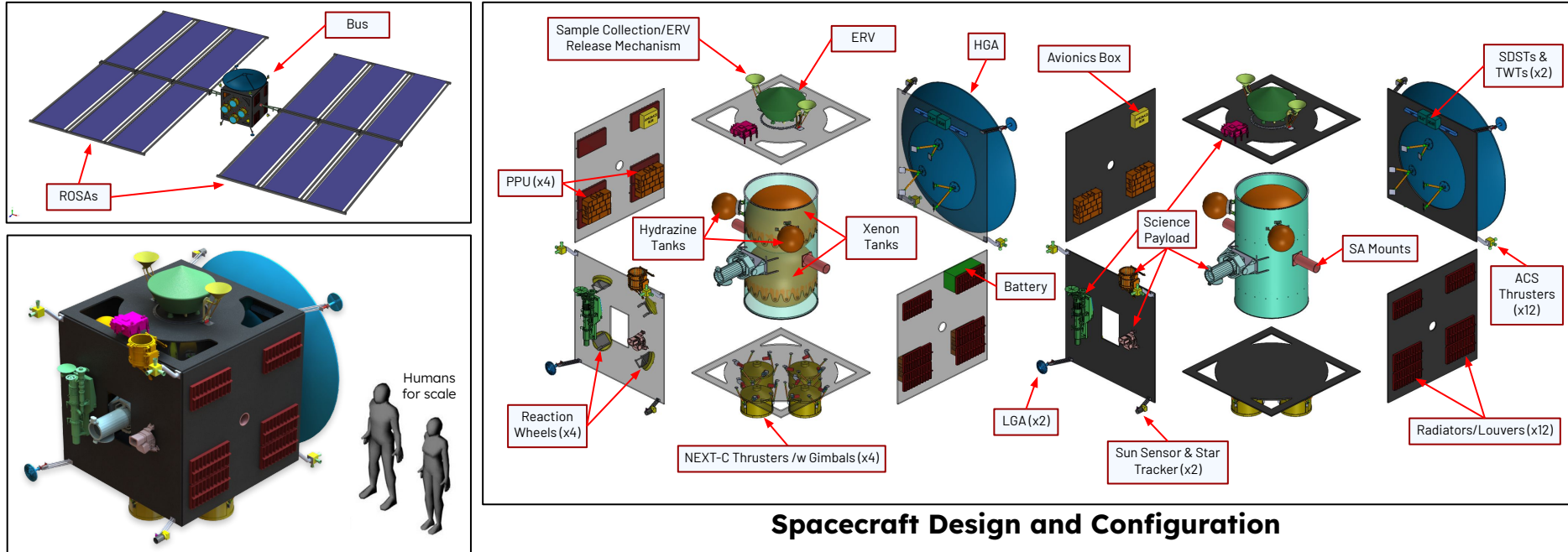


Erik Kramer | Mock Europa Sample Return Mission

Project → A pre-Phase A Europa sample return mission proposal with details for all subsystems

Tasks → Design of **structures** subsystem, **CAD** of all spacecraft components/configurations, **systems engineering** work



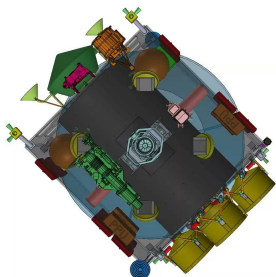
UCLA

- Develop a proposal, including all spacecraft subsystems, to return a water plume sample from Europa
- Launch within 2030-2033 and have a mission duration of 10-12 years

- Used **SolidWorks** to design and model 100+ spacecraft components
- Tracked parts list, mass, and power through linked design spreadsheets, **MEL** and **PEL**
- Assisted other subsystems with **design iterations** and **trade studies**

- Produced a comprehensive SEP mission concept, including spacecraft design, that satisfies all level 1 to 3 requirements and lasts 11.14 years
- Collaborated to produce a 130 page report documenting the design and mission proposal

[Click](#) for Transparent Bus Rotation Animation



[Click](#) for Design Spreadsheet Example

[illegible]

[Click](#) for Solar Array Deployment Animation